

Chapter 3. **GROUPS**

3.2 Subgroups



Today we will learn a method to check whether a subset of a **FINITE** group is a subgroup. It is known as the

FINITE SUBGROUP TEST

FINITE SUBGROUP TEST. If H is a finite nonempty subset of a group G , then H is a subgroup of G if and only if H is closed under the operation of G :

$$h, h_1 \in H \Rightarrow hh_1 \in H, \forall h, h_1 \in H.$$

Proof. $\Rightarrow$ | $\checkmark$

$\Leftarrow$ | Suppose that H is closed under the operation of G . For $h \in H$, the powers $h, h^2, h^3, \dots$ are in H . But since H is finite, we can find two ^{positive} integers such that $h^n = h^{n+m} = h^n \cdot h^m$. Applying the Cancellation Law in G we obtain that $\underline{1 = h^m \in H.}$

$$1 = h \cdot h^{m-1} \Rightarrow \underline{h^{-1} = h^{m-1} \in H}$$

By the Subgroup Test we obtain that H is a subgroup of G . $\blacksquare$

Example 3.17. Consider the cyclic group of order 4: $C_4 = \{1, a, a^2, a^3\}$, where $a^4 = 1$. We can use the *Finite Subgroup Test* to check that $\{1\}$, $\{1, a^2\}$ and C_4 are subgroups of C_4 .

It is obvious that $\{1\}$ and C_4 are subgroups of C_4 . To check that $H = \{1, a^2\}$ is a subgroup of C_4 , using the Finite Subgroup Test, we only need to check that $a^2 \cdot a^2 = a^4 = 1 \in H$ ✓
 $1 \cdot a^2 = a^2 \in H$ ✓

Moreover, $\{1\}$, H and C_4 are the only subgroups of C_4 . Suppose that K is a subgroup of C_4 and $K \neq \{1\}$ and $K \neq \{1, a^2\} = H$. There are two possibilities either $a \in K$ or $a^3 \in K$.

•) If $a \in K$, since K is a subgroup of C_4 we have that $a^2, a^3 \in K$, and

$$\text{so } \underline{K = C_4}.$$

•) If $a^3 \in K$, then

$$(a^3)^2 \in K \text{ and } (a^3)^3 \in K.$$

$$(a^3)^2 = a^6 = a^{1 \cdot 4 + 2} = a^4 \cdot a^2 \stackrel{a^4=1}{=} a^2 \Rightarrow \underline{a^2 \in K}$$

$$\begin{aligned} \underline{(a^3)^3} &= a^3 \cdot (a^3)^2 = a^3 \cdot a^2 = a^5 = a^{1 \cdot 4 + 1} \\ &= a^4 \cdot a = a \stackrel{a^4=1}{=} a \Rightarrow \underline{a \in K} \end{aligned}$$

$$\text{Thus: } \underline{K = C_4}.$$

C_4

$| \subseteq$

$\{1, a^2\}$

$| \subseteq$

$\{1\}$

Lattice
Diagram of
the subgroups
of C_4

Example 3.18. Let us now find all the subgroups of the Klein group:

$$K_4 = \{1, a, b, c\}, \text{ where } a^2 = b^2 = c^2 = 1,$$

and the product of two of a , b , and c is the third.

Using the Finite subgroup test, it is trivial to check that $\{1, a\}$, $\{1, b\}$ and $\{1, c\}$ are subgroups of K_4 . We also have the trivial subgroups: $\{1\}$, K_4 .

If H is a subgroup of K_4 with order ≥ 3 , then H must contain 1 and two of a, b, c , but then H must contain the product of such two (nontrivial) elements, which is the remaining element of K_4 . This shows that $H = K_4$.

